

DESIGN AND IMPLEMENTATION OF AN AUTOMATED WAREHOUSE VENTILATION SYSTEM

Electrical Systems Design Laboratory
EECE 4101

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Submitted by

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How to use this template. Reports are submitted **individually**. Everything already written here is *shared project data* — measurements, tables, derivations, figures — which both authors may use because both produced it. Every *highlighted* block is prose you write yourself, in your own words. Do not swap prose with your partner.

Format is already set to department spec: Times New Roman 12, double spaced, 1.5 in top and left margins, 1 in bottom and right. Cite in square brackets after the last word of the citation [1]. Footnotes are not permitted. Delete this box before submitting.

ABSTRACT

[WRITE — one page, single spaced. State what was built, how, the significant result, and why it matters. Write this LAST. A guide: name the system and its purpose in one sentence; state the control approach in one; give the headline measured result (the system engaged cross-ventilation at 80.8 °F and released at 76.9 °F, and all five control states were verified on hardware); close with the significance and the honest limit.]

TABLE OF CONTENTS

[Generate automatically: References > Table of Contents in Word, or Insert > Table of Contents in LibreOffice. Update it last, after the text is final and page numbers have settled.]

LIST OF FIGURES

[Auto-generate from figure captions. Expected figures: system block diagram; circuit schematic; wiring diagram; state-transition flowchart; enclosure airflow layout; measured cooling curve; measured hysteresis trace. The first five already exist as PDFs in docs/.]

LIST OF TABLES

[Auto-generate. Expected tables: bill of materials with costs; state-transition matrix; pin assignment; measured state verification; air-change rate by chamber volume; power budget.]

NOMENCLATURE

Symbol	Meaning	Unit
$\dot{V}$	Volumetric airflow	m^3/s
$\dot{m}$	Mass flow rate of air	kg/s
V	Chamber internal volume	m^3
ρ	Density of air (1.204 at 20 °C)	kg/m^3
c_p	Specific heat of air at constant pressure (1005)	$\text{J}/(\text{kg}\cdot\text{K})$
$\dot{Q}$	Rate of heat removal	W
Q_s	Internal heat source power	W
τ	Thermal time constant	s
T_i, T_o	Inside and outside air temperature	$^{\circ}\text{C}$
ACH	Air changes per hour	h^{-1}

I. INTRODUCTION

[WRITE — subject, purpose, scope and background. Points to cover, in your own words: warehouses accumulate heat from equipment and solar gain; conventional exhaust fans run on a timer or a single thermostat and therefore ventilate even when outside air is hotter, which adds heat; this project builds a closed-loop controller that decides from BOTH inside and outside temperature; scope is a scaled prototype in a transparent enclosure, thermal control only, humidity deliberately excluded. End the section with one paragraph summarising each section that follows — the department format asks for this.]

II. DESCRIPTION OF THE PROJECT

The system is a closed-loop thermal ventilation controller built on an ESP32-WROOM-32 microcontroller. Two DS18B20 digital temperature probes measure chamber and ambient air on separate one-wire buses. A dual-channel DRV8833 H-bridge drives two 80 mm brushless fans arranged for diagonal cross-ventilation: intake low on one wall, exhaust high on the opposite wall. Three SSD1306 OLED panels display inside temperature, outside temperature and system state. An active buzzer signals state transitions and sensor faults.

[WRITE — expand: the enclosure, how the airflow path is arranged and why diagonal, where the sensors sit and why the ambient probe must stay clear of the exhaust plume. Reference the block diagram figure here.]

III. DESIGN OBJECTIVE

To design and prototype a ventilation system that independently monitors internal and external temperature and dynamically drives a dual-fan intake/exhaust arrangement with no human intervention.

The governing principle is to move air only when the air drawn in is cooler than the air pushed out, and otherwise to stop moving air. Stated as a performance claim: the system drives the chamber toward the lowest temperature obtainable by air exchange, and never takes an action that makes the chamber hotter.

[WRITE — list the specific objectives as numbered items (O1, O2, ...). Suggested: O1 closed-loop control from two independent sensors; O2 fail-safe behaviour on sensor loss; O3 live visualisation of the controlled variables and state; O4 physical demonstration of airflow under a heat source.]

IV. DESIGN REQUIREMENTS

The following specifications and constraints were fixed during design. Each is justified analytically in Section V.

Requirement	Specification	Rationale
Threshold hysteresis	ON at 80.0 °F, OFF at 77.0 °F	A bare threshold chatters on sensor noise; band exceeds twice the ± 0.90 °F sensor tolerance
Differential deadband	Enter at 2.0 °F cooler, exit at 1.0 °F	Two probes carry ± 1.80 °F of worst-case differential uncertainty
Minimum state dwell	10 s	Four sensor periods; a state cannot change on a single reading
Sensor fault threshold	3 consecutive bad reads	Tolerates one glitch while still responding within 7.5 s
Failure behaviour	Explicit FAULT state, audible, exhaust driven	A missing reading must never render as a plausible value or default silently to idle
Logic levels	3.3 V throughout, no level shifter	Every peripheral selected 3.3 V native; removes the most dangerous wiring error the design allowed
Motor supply	Separate 5 V 2 A, common ground	Sized by fan inrush, not steady draw; prevents brownout of the controller
Control loop	Non-blocking, cooperative	A DS18B20 conversion takes up to 750

scheduler, no delay() in loop ms; the loop must never wait on a sensor

Control units Celsius internally, converted Sensor native units; avoids repeated
only for display conversion error

V. ALTERNATIVE SOLUTIONS AND DESIGN PROCEDURE

A. Alternatives considered

Decision	Alternative rejected	Basis for the choice
Brushless computer fans	130-size brushed motors with propellers	Brush noise corrupts the I ² C bus, the dominant failure mode of the earlier design; also quieter, better balanced, and simpler to mount
		The LCD is 5 V and would require a level shifter beside a 3.3 V controller; each measured value also gets a full panel, readable across a room
Three OLED panels	Single 16×2 HD44780 LCD	
	Reversing the intake fan to block backdraft	Ventilating while outside is hotter adds heat; stopping both fans solves it with no reversing hardware
Thermal-only control	Include humidity sensing	Streamlines the control loop and removes a failure mode with no benefit to the stated objective
		Fans switch only on and off so an H-bridge is not required, but the DRV8833 needed no wiring change and its reverse capability assisted bring-up
DRV8833 H-bridge	Two logic-level N-MOSFETs	

[WRITE — the syllabus and course outcome SLO-2 require alternatives to be evaluated]

on social, economic and societal impact as well as technical merit. Add a paragraph: energy use against always-on ventilation, commodity component cost and availability, noise in an occupied space, and safety of a low-voltage design.]

B. Control design

Control is implemented as a five-state finite state machine evaluated once per sensor period.

State	Entry condition	Intake	Exhaust	Alarm
STANDBY	T_{in} below 80 °F	off	off	—
CROSS_VENT	T_{in} above 80 °F and outside at least 2 °F cooler	on	on	—
EXHAUST_ONLY	T_{in} above 80 °F and outside within 3 °F	off	on	chirp on entry
SEALED	T_{in} above 80 °F and outside at least 3 °F hotter	off	off	chirp on entry
FAULT	3 consecutive bad sensor reads	off	on	continuous tone

[WRITE — justify each transition. Why EXHAUST_ONLY rather than sealing when temperatures are close; why FAULT drives the exhaust rather than stopping (a bounded chamber near ambient beats an unbounded one); why SEALED cannot deadlock. Reference the state-transition flowchart figure.]

C. Analytical design calculations

Airflow per fan, from the 20 CFM rating:

$$\dot{V} = 20 \times 4.7195 \times 10^{-4} = 9.439 \times 10^{-3} \text{ m}^3/\text{s}$$

Mass flow and the rate of sensible heat removal:

$$\dot{m} = \rho \dot{V} = 1.204 \times 9.439 \times 10^{-3} = 1.136 \times 10^{-2} \text{ kg/s}$$

$$\dot{Q} = \dot{m} c_p (T_i - T_o) = 11.4 \cdot \Delta T \text{ [W]}$$

The system therefore removes approximately 11.4 W per kelvin of inside-to-outside difference, or about 57 W at $\Delta T = 5$ K. The same expression is negative when outside is hotter: ventilating under that condition adds heat at the same rate, which is the analytical basis for the SEALED state.

An energy balance on a well-mixed chamber with an internal source gives

$$V \rho c_p (dT_i/dt) = \dot{m} c_p (T_o - T_i) + Q_s$$

from which the steady state is

$$T_{i,ss} = T_o + Q_s / (\dot{m} c_p)$$

With no internal source the chamber settles at ambient exactly. The system contains no refrigeration and therefore cannot cool below ambient; with a source it settles approximately 1 K above ambient per 11.4 W. Increasing airflow reduces that elevation but cannot make it negative.

The homogeneous solution is first order with

$$T_i(t) = T_o + (T_{i,0} - T_o)e^{-t/\tau}, \quad \tau = V / \dot{V}$$

Mass and specific heat cancel, so the thermal time constant is simply the chamber volume divided by the volumetric flow rate — one air-change time.

Chamber volume V (m³) τ free-air (s) ACH τ at 60% derate (s) ACH

12 gal	0.0454	4.8	748	8.0	449
18 gal	0.0681	7.2	499	12.0	299
27 gal	0.1022	10.8	332	18.0	199
32 gal	0.1211	12.8	281	21.4	168

[REPLACE this table with the single row for the measured chamber volume once the enclosure is built.]

D. Derivation of the control thresholds

The DS18B20 is specified to $\pm 0.5\text{ }^{\circ}\text{C}$, i.e. $\pm 0.90\text{ }^{\circ}\text{F}$. Quantisation at 12-bit resolution is $0.0625\text{ }^{\circ}\text{C}$, an order of magnitude smaller, so tolerance dominates. To prevent re-triggering on noise the hysteresis band must exceed the peak-to-peak uncertainty of one sensor:

$$\text{band} > 2 \times 0.90 = 1.80\text{ }^{\circ}\text{F} \rightarrow 80.0/77.0\text{ }^{\circ}\text{F} \text{ gives } 3.00\text{ }^{\circ}\text{F}, \text{ a margin of } 1.67$$

The differential comparison involves both probes. Worst case the tolerances add to $\pm 1.80\text{ }^{\circ}\text{F}$; treated as independent the root-sum-square figure is $\pm 1.27\text{ }^{\circ}\text{F}$. The worst case was adopted because both probes are the same part from the same batch and their errors may be correlated rather than cancelling. The $2.0\text{ }^{\circ}\text{F}$ entry deadband therefore clears the worst-case uncertainty, and exit at $1.0\text{ }^{\circ}\text{F}$ provides differential hysteresis.

E. Power budget

Load	Current	Rail	Power
$2 \times 80\text{ mm fan, steady}$	200 mA	5 V	1.00 W
ESP32-WROOM-32, radio unused	80 mA	3.3 V	0.26 W
$3 \times \text{SSD1306 OLED}$	60 mA	3.3 V	0.20 W
$2 \times \text{DS18B20}$	3 mA	3.3 V	0.01 W
Buzzer module	30 mA	3.3 V	0.10 W
DRV8833 quiescent	2 mA	5 V	0.01 W
Total steady			1.58 W

The supply is sized by inrush rather than steady draw. A brushless fan draws roughly three times its rated current while the rotor accelerates, and both fans start together on entry to CROSS_VENT, giving a worst case of approximately 0.6 A. The 5 V 2 A supply provides 3.3 $\times$ headroom on that transient.

VI. PRESENTATION OF DESIGN TESTING RESULTS

A. Verification of all control states

Each state was driven on assembled hardware and the commanded outputs recorded from the controller trace.

Inside / outside (°F)	State reached	Intake	Exhaust	Matches design
72 / 65	STANDBY	off	off	yes
85 / 65	CROSS_VENT	on	on	yes
85 / 84	EXHAUST_ONLY	off	on	yes
85 / 95	SEALED	off	off	yes
85 / 65 (returning)	CROSS_VENT	on	on	yes — seal released unaided
78 / 65	CROSS_VENT	on	on	yes — hysteresis holds
74 / 65	STANDBY	off	off	yes

B. Closed-loop response to a real heat source

With both probes live and no injected values, the chamber probe was warmed by hand and released.

Inside temperature	State	Fans
75.2 °F	STANDBY	off
80.8 °F rising	CROSS_VENT	both on
87.1 °F	CROSS_VENT	both on
77.9 °F falling	CROSS_VENT	both on (latch holds)
76.9 °F falling	STANDBY	off

Engagement at 80.8 °F and release at 76.9 °F confirm the designed hysteresis operating against real sensor noise. The outside probe moved −0.1 °F over the same interval while the inside probe moved +11.8 °F, confirming that the two channels are independent and correctly assigned.

C. Fault behaviour

The chamber probe's data line was disconnected during operation and reconnected.

Required behaviour	Observed
A missing reading never renders as a plausible value	displayed as "--.-", not 0.0
Several bad reads required, so one glitch cannot trip it	exactly 3 consecutive
Fails safe to exhaust driven, never sealed	exhaust driven
Audible and distinguishable from a state change	continuous tone
Recovers without intervention	returned to STANDBY, counter reset

[WRITE — a short paragraph introducing these three tests and what each was intended to establish. Reference the measured traces as figures.]

VII. DATA ANALYSIS

[This section requires the enclosure and the heat-source runs. Do not write it until the data exists.]

[WRITE — procedure: heat the chamber to a steady elevated temperature, cut the source, allow the controller to ventilate, and log inside temperature against time.]

Fitting the logged curve to the first-order model, a plot of

$$\ln[(T_i(t) - T_o) / (T_{i,0} - T_o)] \text{ versus } t$$

is a straight line of slope $-1/\tau$, giving the measured time constant. The ratio of the measured value to the predicted $V/\dot{V}$ is the installed flow derate, so a single experiment yields both the thermal response and the effective airflow.

Quantity	Predicted Measured Ratio		
Time constant τ (s)	$[V/\dot{V}]$	$[from\ fit]$	$[]$
Air changes per hour	$[3600/\tau]$	$[]$	$[]$
Steady-state elevation above ambient (K)	$[Q_s/\dot{m}c_p]$	$[]$	$[]$

[WRITE — interpret. Does the measured derate agree with the 50–70% expected for an installed fan? Which assumption in Section V dominates the discrepancy? Compare the sealed case against the ventilated case over the same interval to quantify what the control law actually achieves.]

VIII. COST ANALYSIS

[Required by course outcome SLO-5. Complete the unit prices from the purchase records.]

Item	Qty	Unit cost	Extended
ESP32-WROOM-32 development board	1	[]	[]
DS18B20 temperature probe module	2	[]	[]
SSD1306 128×64 OLED panel	3	[]	[]
DRV8833 dual motor driver	1	[]	[]
80 mm 5 V brushless fan	2	[]	[]
Active buzzer module	1	[]	[]
5 V 2 A power supply	1	[]	[]
Transparent enclosure, fasteners, weatherstrip	1	[]	[]
Breadboards, jumper wire, consumables	—	[]	[]
Total			[]

[WRITE — comment on cost relative to function: the controller and sensing account for a small fraction of the total, and the design deliberately avoids parts that would raise cost without improving the result. Note that scaling to a real warehouse changes the fan and supply selection but not the control law.]

IX. CONCLUSION

[WRITE — a precise statement of results answering each objective from Section III in turn. Cover: the controller operates closed-loop from two independent sensors with no human input; all five states were verified on assembled hardware; the hysteresis was confirmed against real sensor noise; the fault path was verified by physically disconnecting a sensor and recovered unaided. State the physical limit honestly — no refrigeration, so ambient is the floor — and give recommendations for a next revision.]

REFERENCES

- [1] Analog Devices/Maxim, “DS18B20 Programmable Resolution 1-Wire Digital Thermometer,” datasheet.
- [2] Texas Instruments, “DRV8833 Dual H-Bridge Motor Driver,” datasheet.
- [3] Solomon Systech, “SSD1306 128×64 Dot Matrix OLED Driver,” datasheet.
- [4] Espressif Systems, “ESP32-WROOM-32 Datasheet.”
- [5] Add any text or standard cited for the thermodynamics.

APPENDICES

Appendix A. Controller firmware source listing.

Appendix B. Automated test suite and results.

Appendix C. Sample calculations.

Appendix D. Pin assignment and wiring schedule.

[Paste from firmware/src/, test/, docs/ and hardware/01_pinout.md.]